We initially had the map $z \mapsto \overline{z}$ which is not holomorphic. Let $f(z) = \overline{z}$ and $\frac{\partial f}{\partial \overline{z}} = 1$ then we've captured this with $\frac{\partial}{\partial \overline{z}}$.
If we have $z, \overline{z}$ as independent variables with $z = x + iy$. Consider the map

$$(z, \overline{z}) \mapsto (x(z, \overline{z}), y(z, \overline{z})) \mapsto f(x, y).$$

We have

$$\partial_z f$$

3 Jan 16 — Power Series and Analytic Functions

3.1 Power Series

Definition 3.1. *Given a $\mathbb{C}$ valued seq $(a_n)_{n=0}^{\infty}$, form*

$$\sum_{n=0}^{\infty} a_n z^n.$$

A power series

1. *Converges if $\lim_{N\to\infty} \sum_{n=0}^{N} a_n z^n$ exists*
2. *Converges absolutely if $\sum_{n=0}^{\infty} |a_n||z|^n$ converges*

Remark 3.2. *By translation, all we say will apply to $\sum a_n (z - z_0)^n$*

Theorem 3.3. *Given $\sum_{n=0}^{\infty} a_n z^n$, $\exists R \in [0, +\infty]$ such that*

1. *It converges, if $D_R = \{|z| < R\}$*
2. *It diverges at every at z such that $|z| > R$.*

Moreover,

$$R = \frac{1}{\lim_{n\to\infty} \sup |a_n|^{\frac{1}{n}}}. \qquad \text{(Hadamard's Formula)}$$

Remark 3.4. *On ∂D_R it can often be delicate.*

Proof. Fix $|z| < R$, then $\frac{1}{R}|z| < 1$ and fix $\epsilon > 0$ such that

$$\left(\frac{1}{R} + \epsilon\right)|z| < 1.$$

Then, eventually

$$|a$$

Theorem 3.6. *The function $f(z) = \sum a_n z^n$ is holomorphic on D_R, f' is a power series*

$$f'(z) = \sum n a_n z^{n-1}.$$

Whose disk of convergence is R.

Proof. Let $g(z) = \sum n a_n z^{n-1}$, $z \in D_R$. We want to show that f' exist and $f' = g$.
Fix $z_0 \in D_R$, let $r < R$ such that $|z_0| < r < R$.
Fix $N \geq 1$ (to be chosen later)

$$f(z) = \sum_{n=0}^{N} a_n z^n + \sum_{n=N+1}^{\infty} a_n z^n = S_N(z) + E_N(z).$$

From here we want to show that

$$\frac{f(z_0+h) - f(z_0)}{h} \to g(z_0).$$

For all h such that $|z_0 + h| < r$

$$\begin{aligned}\frac{f(z_0+h) - f(z_0)}{h} - g(z_0) &= \left(\frac{S_N(z_0+h) - S_N(z_0)}{h} - S_N'(z_0)\right) + (S_N'(z_0) - g(z_0)) + \left(\frac{E_N(z_0+h) - E_N(z_0)}{h}\right) \\ &= A + B + C\end{aligned}$$

Fix $\epsilon > 0$ we want to show that $\exists \delta$ such that $\forall |h| < \delta, |A + B + C| < \epsilon$.

$$\begin{aligned}|C| &\leq \sum_{n=N+1}^{\infty} |a_n| \left|\frac{(z_0+h)^n - z_0^n}{h}\right| \\ &= \sum_{n=N+1}^{\infty} |a_n| \left|(z_0+h)^{n-1} + (z_0+h)^{n-2} z_0 + \cdots + z_0^n\right| \\ &\leq \sum_{n=N+1}^{\infty} |a_n||nr^{n-1}| < \frac{\epsilon}{3} \qquad \text{(for all } N \geq N_1(\epsilon))\end{aligned}$$

Note the last summation is the tail of $\sum a_n n r^{n-1}$ which converges by $r < R$.
For B:

$$S_N'(z_0) = \sum_{n=0}^{N} n a_n z_0^{n-1} \to g(z_0). \qquad \text{(by def of } g)$$

So $\forall N \geq N_2$, $|B| < \frac{\epsilon}{3}$.
$|A| < \frac{\epsilon}{3}$ for all h sufficiently small by the definition of the complex derivative and the fact that S_N is

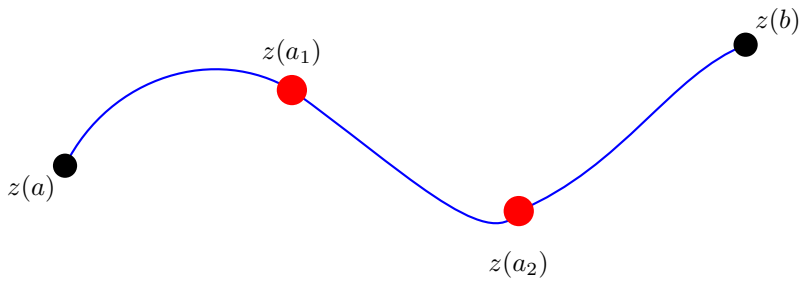
4 Jan 21 — Parametrized Curves and Contour Integration

4.1 Parametrized Curves

Definition 4.1. *A parametrized curve (in $\mathbb{C}$) is a function $z(t) : [a, b] \to \mathbb{C}$*

1. *Smooth if z' is continuous and $\forall t \in [a, b]$, $z'(t) \neq 0$*
2. *Piecewise smooth if z is continuous and there exists $a_0 = a < a_1 < \cdots < a_k = b$ such that $z|_{[a_j, a_{j+1}]}$ is smooth for all $j = 0, \ldots, k-1$.*
3. *Let $\tilde{z} : [\tilde{a}, \tilde{b}] \to \mathbb{C}$ then two parameterization are equivalent if there exist c^1-bijection $s \mapsto t(s) : [\tilde{a}, \tilde{b}] \to [a, b]$ and $t'(s) > 0$ for all s such that $\tilde{z}(s) = z(t(s))$.*

Example 4.2. *A piecewise smooth curve can be visualized as follows:*



The curve consists of smooth segments joined at corner points (marked in red) where the derivative is discontinuous.

Definition 4.3. *The family of all equivalent parameterization determines a curve (oriented) $\gamma \subset \mathbb{C}, \gamma = z([a, b])$.*
$z(a)$ is the starting point and $z(b)$ is the endpoint of γ. $\gamma^{-1} = \gamma$ with reversed orientation

Definition 4.4. *γ is closed if $z(a) = z(b)$ and simple if $z(t_1) \neq z(t_2)$ for all $t_1 \neq t_2$.*

Example 4.5. $z(T) = re^{-it}, t \in [0, 2\pi]$

Definition 4.9. *For a piece-wise smooth curve, we define*

$$\int_\gamma f = \sum_{j=0}^{k-1} \int_{a_j}^{a_{j+1}} f(z(t))z'(t)dt.$$

Definition 4.10. *We define*

$$|\gamma| = \mathit{length}(\gamma) = \int_a^b |z'(t)|dt.$$

Theorem 4.11. *Some very basic properties:*

1. *Linearity*
$$\int_\gamma (\alpha f + \beta g) = \alpha \int_\gamma f + \beta \int_\gamma g.$$
2.
$$\int_{\gamma^{-1}} f = -\int_\gamma f.$$
3.
$$\left| \int_\gamma f \right| \leq \left(\sup_\gamma |f| \right) |\gamma|.$$

4.3 Primitives and the Fundamental Theorem

Definition 4.12. *The primitive of $f : \Omega \to \mathbb{C}$ with open $\Omega \subset \mathbb{C}$ is any $F : \Omega \to \mathbb{C}$ holomorphic such that $F' = f$ on Ω*

Theorem 4.13. *The Fundamental Theorem of Calculus*
For f with primitive F on Ω, $\gamma \subset \mathbb{C}$ curve from $w_1 \to w_2$ we have

$$\int_\gamma f = F(w_2) - F(w_1).$$

Proof. We can write

$$\int_\gamma f = \int_a^b f(z(t))z'(t)dt = \int_a^b \frac{d}{dt}(F(z(t)))dt = F(z(b)) - F(z(a)).$$

□

Corollary 4.14. *If γ is closed then*

$$\int_\gamma f = 0.$$

Corollary 4.15. *If f is holomorphic on Ω*

5 Jan 23 — Elementary Functions and Fundamental Theorem of Algebra

5.1 Glossary of Elementary Functions

Glossary of Elementary Functions

1. Polynomials: $P(z) = a_0 + a_1 z + \cdots + a_n z^n$ for $a_i \in \mathbb{C}$ and is entirely holomorphic in $\mathbb{C}$, $a_n \neq 0$ and $n = \deg P$
2. Rational: $R(z) = \frac{P(z)}{Q(z)}$ where P, Q are polynomials with no common factors. The zeros of Q are called poles and their order is called the order of a given pole.
3. Exponential: $e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$

5.2 Fundamental Theorem of Algebra

Lemma 5.1. *Let $P(z)$ be a complex polynomial of positive degree n, if $|P(z)|$ has a local minimum at z_0 then $P(z_0) = 0$.*

Proof. WLOG $z_0 = 0$ as we can translate $Q(z) = P(z - z_0)$.
AFSOC $P(0) \neq 0$ and WLOG $P(0) = 1$ by $\frac{P}{P(0)}$. We can write $P(z) = 1 + a_k z^k + \cdots + a_n z^n$ with $a_k, a_n \neq 0$.
Consider $z_\delta = \delta e^{i\theta}$. For all $\delta > 0$ sufficiently small:

$$\left|a_{k+1} z_\delta + \cdots + a_n z_\delta^{n-k}\right| < \frac{|a_k|}{2}.$$

Then

$$|P(z)| \leq \left|1 + a_k z_\delta^k\right| + \left|a_{k+1} z_\delta^{k+1} + \cdots + a_n z_\delta^n\right| < \left|1 + a_k z_\delta^k\right| + \left|z_\delta^k\right| \frac{|a_k|}{2}.$$

Now we can choose δ such that $a_k z_\delta^k < 0$ and real.

$$\left|1 + a_k z_\delta^k\right| + \left|z_\delta^k\right| \frac{|a_k|}{2} = 1 + a_k z_\delta^k - \frac{1}{2} a_k z_\delta^k = 1 + \frac{1}{2} a_k z_\delta^k < 1.$$

□

Theorem 5.2. *Fundamental Theorem of Algebra:*

Proof. $|P(z)| \to +\infty$ as $|z| \to +\infty$ therefore $\exists R$ such that $\forall |z| > R$, $|P(z)| > |P(0)|$ thus $\inf_{z \in \mathbb{C}} |P(z)| = \inf_{z \in D_R} |P(z)|$ and such a value is attained somewhere at z_0 then by lemma 5.1 we're done. □

Lemma 5.3. *$P(z) = a_n (z - z_1) \cdots (z - z_n)$ where $z_1, ..., z_n$ are roots of P.*
The order of a zero z_j is its multiplicity, the number of times it appears in the sequence. If z_j is of the order k_j,

$$P(z) = Q(z)$$

Theorem 5.6. The following are true

1. $(e^z)^0 = e^z$
2. $e^{w+z} = e^w e^z$
3. $\overline{e^z} = e^{\overline{z}}$
4. $je^{iy}j = 1$ and $je^{x+iy}j = e^x$
5. $\sin(z) = \frac{e^{iz} - e^{-iz}}{2i}$ and $\cos(z) = \frac{e^{iz} + e^{-iz}}{2}$

6 Jan 26 | Cauchy's and Goursat's Theorems

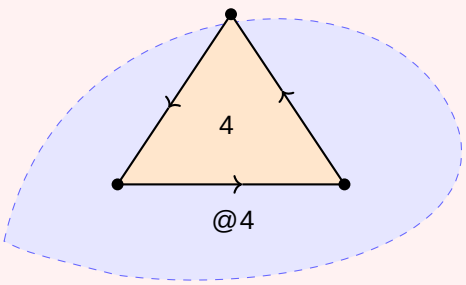
6.1 Cauchy's and Goursat's Theorems

Theorem 6.1. Cauchy Theorem: If $f : \; ! \; \mathbb{C}$ is holomorphic on $\quad$ and $\quad$ is a closed curve in $\quad$ then

$$\int_{Z} f = 0:$$

Theorem 6.2. Goursat's Theorem: If $f : \; ! \; \mathbb{C}$ is holomorphic and $\quad$ is open, then 84 $\quad$ we have

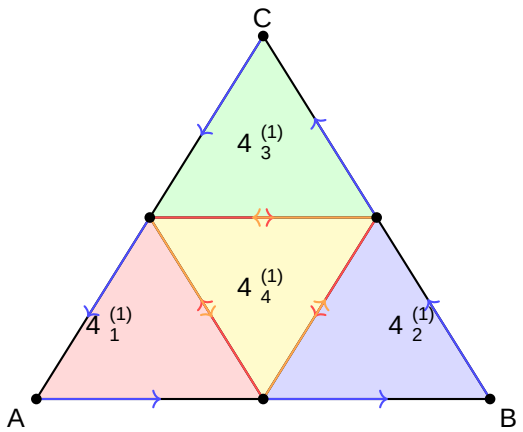
$$\int_{@4} f = 0:$$



The integral around the positively oriented (counterclockwise) boundary of any triangle 4 $\quad$ vanishes.

Proof. Let $4 = 4^{(0)}$ and partition into four triangles by connecting the midpoints of the sides:

$$4^{(0)} = 4^{(1)}_1 \; [\; 4^{(1)}_2 \; [\; 4^{(1)}_3 \; [\; 4^{(1)}_4 :$$



13 Feb 13 — Classification of Singularities

13.1 Classification of Singularities

Theorem 13.1. *(Riemann) Let $f : \Omega \setminus \{z_0\} \to \mathbb{C}$ has a removable singularity at z_0 if and only if bounded near z_0.*

Corollary 13.2. *Pole at z_0 if and only if $|f| \to +\infty$ as $z \to z_0$*

Proof. ($\implies$) Local description near poles implies $f(z) = (z - z_0)^{-n} g(z)$ so $|f| \to +\infty$
($\impliedby$) Since $\left|\frac{1}{f}\right| \to 0$, in particular $\frac{1}{f}$ is bounded near z_0, Riemann implies $\frac{1}{f}$ has a removable singularity at z_0. □

Theorem 13.3. *(Casorati-Weierstrass) If $f : D_r(z_0) \setminus \{z_0\} \to \mathbb{C}$ is holomorphic with an essential singularity at z_0 then the image of f is dense in $\mathbb{C}$.*

Proof. If not, $\exists w \in \mathbb{C}$ such that no values of f near w, $\exists \delta > 0$ s.t. $\forall z$, $|f(z) - w| > \delta$.
Consider

$$g(z) = \frac{1}{f(z) - w} : D_r(z_0) \setminus \{z_0\} \to \mathbb{C}.$$

Such a g is bounded by $\frac{1}{\delta}$. So g extends halomorphically into $D_r(z_0)$.

(Case 1) $g(z_0) = 0 \implies \frac{1}{g} \to +\infty$ so $f(z) - w$ has a pole so f has a pole

(Case 2) $g(z_0) \neq 0 \implies \frac{1}{g}$ is well defined and f has a removable singularity.

□

Theorem 13.4. *(Picard's Theorem) If $f : D_r(z_0) \setminus \{z_0\} \to \mathbb{C}$ is holomorphic with an essential singularity at z_0 then the image of f is either $\mathbb{C}$ or $\mathbb{C}$ except for at most one value.*

13.2 Meromorphic Functions and Behavior at Infinity

Definition 13.5. *Let $\Omega \subset \mathbb{C}$ be open. A function f is called meromorphic on Ω if $\exists z_1, z_2, \ldots$ without a limit point such that f is holomorphic in $\Omega \setminus \{z_1, z_2, \ldots\}$ and has poles at $z_1, z_2, \ldots$.*

Theorem 13.6. *(Behavior at $+\infty$) Suppose f is holomorphic in a neighborhood of $_\infty$ i.e. $\{|z| > R\}$ for some $R > 0$.*
Note: $F(z) = f(\frac{1}{z})$ is holomorphic in $D_{\frac{1}{R}}(0) \setminus \{0\}$.
We say f has a removable singularity at ∞ if F has a removable singularity at $z = 0$.

Example 13.7. *e^z has an essential singularity at $+\infty$.*

Remark 13.8. *We'll denote $\overline{\mathbb{C}} = \mathbb{C} \cup \{+\infty\}$*

Example 13.9. *Examples of meromorphic functions on $\mathbb{C}$*

1. *Rational functions: $\frac{P}{Q}$*
2. *The gamma function: $\Gamma(z) = \int_0^\infty e^{-t} t^{z-1} dt$ (simple poles at $\ldots, -2, -1, 0$)*
3. *The Riemann zeta function: $\zeta(s) = \sum_{n=1}^\infty \frac{1}{n^s}$ (simple poles at $s = -1, -2, -3, \ldots$)*

14 Feb 16 — Rational Functions via Meromorphy

14.1 Rational Functions via Meromorphy

Theorem 14.1. *Suppose f is meromorphic on $\mathbb{C}$ with a removable singularity or a pole at $+\infty$ then f is a rational function.*

Proof. Claim: Under these assumptions, f has finitely many poles.
If not, $z_1, z_2, \ldots$ are poles and this sequence cannot be bounded otherwise it got a limit point. Say $|z_k| \to +\infty$, the neighborhoods of ∞ are $\{|z| > R\}$ so f not holomorphic for any $R > 0$ which contradicts that f has a removable singularity at $+\infty$.

For a function F which has a pole at $z = 0$ we have

$$F(z) = \underbrace{\frac{a_n}{z^n} + \ldots + \frac{a_{-1}}{z}}_{f(z)} + \underbrace{a_0 + a_1 z + \cdots}_{g(z)}.$$

Let $z_1, \ldots, z_N$ be the poles of f. Near z_k, for each $1 \leq k \leq N$,

$$f(z) = \underbrace{f_k(z)}_{\substack{\text{principal part} \\ \text{poly in } \frac{1}{z-z_k}}} + \underbrace{g_k(z)}_{\text{holomorphic}}.$$

Near ∞,

$$f\left(\frac{1}{z}\right) = \underbrace{\tilde{f}_\infty(z)}_{\substack{\text{principal part} \\ \text{poly in } \frac{1}{z}}} + \underbrace{\tilde{g}_\infty(z)}_{\text{holomorphic}}.$$

Consider,

$$H(z) = f(z) - \underbrace{\sum_{j=1}^{N} f_j(z)}_{\text{rational}} - \underbrace{\tilde{f}_\infty(z)}_{\text{poly}}.$$

Claim: H is holomorphic and bounded.
Near

15 Feb 18 — Argument Principle and Maximum Modulus

15.1 Argument Principle and Rouche's Theorem

Theorem 15.1. *(The Argument Principle) Let $f : \Omega \to \mathbb{C}$ be meromorphic, $\overline{D} \subset \Omega$ be a disk. If If f never vanishes on ∂D nor has nay poles on ∂D then*

$$\frac{1}{2\pi i}\int_{\partial D}\frac{f'}{f} = \#\text{ zeros of } f - \#\text{ poles of } f \text{ in } \overline{D}.$$

Proof. If f has zero of order n at z_0 then $f(z) = (z - z_0)^n g(z)$ nonvanishing near z_0 then

$$\frac{f'}{f}(z) = \frac{n}{z - z_0} + \frac{g'}{g}(z).$$

If f has a pole of order n at z_0 we can write $f(z) = (z - z_0)^{-n} h(z)$ then

$$\frac{f'}{f}(z) = -\frac{n}{z - z_0} + \frac{h'}{h}(z).$$

Therefore $\frac{f'}{f}$ has simple poles at zeros and poles of f with residue $\frac{\text{order}}{-\text{order}}$. The residue formula finishes the proof. □

Theorem 15.2. *Rouche's Theorem: Let f, g be holomorphic in $\Omega \supset \overline{D}$ with Ω open. If $|f| > |g|$ on ∂D then $f, f + g$ have the same number of zeros in D.*

Proof. Let $f_t = f + t \cdot g$ for $t \in [0, 1]$ and $n_t = \#$ of zeros of f_t in D. By argument principle we have

$$n_t = \frac{1}{2\pi i}\int_{\partial D}\frac{f_t'}{f_t}.$$

f_t never vanishes on ∂D. We can conclude

$$|f_t| \geq |f| - t|g| \geq |f| - |g| > 0.$$

We can show continuity of the integrand by

$$\int_0^{2\pi} \underbrace{\frac{f_t'(z_0 + re^{i\theta})}{f_t(z_0 + re^{i\theta})} rie^{i\theta}}_{H(t,\theta)}\, d\theta.$$

Then $H(t, \theta)$ is jointly continuous in (t, θ). So n_t is continuous and discrete so n_t is constant. □

Proof. Third Proof of fundamental Theorem of Algebra.

$$P(z) = \underbrace{a_0 + a_1 z + \cdots + a_{n-1} z^{n-1}}_{g(z)} + \underbrace{a_n z^n}_{f(z)}.$$

There exist $\mathbb{R} > 0$ such that $|g(z)| < |f(z)|$ on $|z| = R$ □

15.2 Open Mapping and Maximum Modulus Theorems

Theorem 15.3. *(The Open Mapping Theorem) Let $f : \Omega \to \mathbb{C}$ holomorphic, f non-constant then f maps open sets to open sets.*

Proof. Let $w_0 = f(z_0) \in f(\Omega)$. We want to show that every w near w_0 is also in $f(\Omega)$ Fix w near w_0 then consider,

$$h(z) = f(z) - w = \underbrace{(f(z) - w_0)}_{F(z)} + \underbrace{(w_0 - w)}_{G(z)}.$$

Let $\delta > 0$ be such that $\overline{D}_\delta(z_0) \subset \Omega$ and $\forall z \in \partial D_\delta(z_0)$, $f(z) \neq w_0$. Otherwise if there exist no such δ then there exist $(z_n) \to z_0$ such that $f(z_n) = w_0$ for all n then by rigidity f is constant which contradicts the assumption that f is non-constant. Let

$$\epsilon = \inf_{\partial D_\delta(z_0)} |f(z) - w_0| > 0.$$

Now $|G| > |F|$ on $\partial D_\delta(z_0)$ as long as $|w - w_0| < \epsilon$. By Rouche's Theorem, $h = F + G$ has a zero in $\overline{D}_\delta(z_0)$. $\square$

Theorem 15.4. *(The Maximum Modulus The1p4 Tf -2 2(G) T /F2 . 2 2 Tf 35.52 4(1 51(i5 4 Tdn*

16 Feb 20 — Homotopy and Simply Connected Regions

16.1 Homotopy and Path Independence

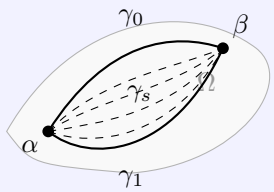
Definition 16.1. *Let $\Omega \subset \mathbb{C}$ be a region, two curves in Ω say with parametrization $\gamma_0, \gamma_1 : [a,b] \to \mathbb{C}$ and common endpoints*

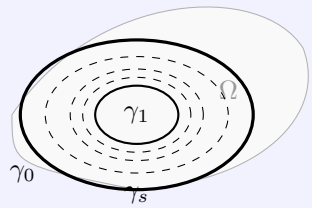
$$\alpha = \gamma_0(a) = \gamma_1(a) \text{ and } \beta = \gamma_0(b) = \gamma_1(b).$$

are called homotopic if $\forall s \in [0,1]$, $\exists \gamma_s : [a,b] \to \Omega$ such that

1. *$\gamma_{s=0} = \gamma_0$ and $\gamma_{s=1} = \gamma_1$,*
2. *$\gamma_s(a) = \alpha$ and $\gamma_s(b) = \beta$ for all $s \in [0,1]$,*
3. *the map $(s,t) \mapsto \gamma_s(t)$ is continuous on $[0,1] \times [a,b]$.*

The map $H(s,t) = \gamma_s(t)$ is called a homotopy between γ_0 and γ_1.





Theorem 16.2. *If $f : \Omega \to \mathbb{C}$ be holomorphic, γ_0, γ_1 are two homotopic curves in Ω then $\int_{\gamma_0} f = \int_{\gamma_1} f$*

Proof. Let $H(s,t) : [0,1] \times [a,b] \to \Omega$ be a homotopy between γ_0, γ_1. Since $K = H([0,1] \times [a,b])$ is compact in open Ω, $\exists \epsilon > 0$ such that $\forall z \in K$, $D_{3\epsilon}(z) \subset \Omega$. Fix such $\epsilon > 0$ and take $\delta > 0$ such that

$$|H(s_1,t_1) - H(s_2,t_2)| < \epsilon \text{ provided that } |(s_1,t_1) - (s_2,t_2)| < 2\delta. \qquad \text{(unif cont of } H\text{)}$$

Given such δ, it suffices to show

$$\int_{\gamma_s} f = \int_{\gamma_{s'}} f \text{ for all } 0 \le s < s' \le 1 \text{ and } |s - s'| < \delta.$$

Take $a = t_0 < t_1 < \cdots < t_n < t_{n+1} = b$ such that $|t_{i+1} - t_i| < \delta$ for all $i = 0, 1, \cdots, n$.
Let's denote $z_i = \gamma_s(t_i)$ and $w_i = \gamma_{s'}(t_i)$. Let $D_i = D_{z_i}(2\epsilon)$.

Claim 16.3.

1. *$z_i, z_{i+1} \in D_i$ holds because $|z_{i+1} - z_i| < \epsilon$ as $z_{i+1} = \gamma_s(t_{i+1}) = H(s, t_{i+1})$ and $z_i = \gamma_s(t_i) = H(s, t_i)$.*
2. *$w_i, w_{i+1} \in D_i$ holds because $|w_i - z_i| < \epsilon$ so*
$$|w_{i+1} - z_i| \le |w_{i+1} - z_{i+1}| + |z_{i+1} - z_i| < \epsilon + \epsilon = 2\epsilon.$$
3. *$D_0 \cup D_1 \cup \cdots \cup D_n$ covers $\gamma_s, \gamma_{s'}$ because for $z = \gamma_s(t)$, $t_i \le t \le t_{i+1}$, $|z - z_i| = |\gamma_s(t) - \gamma_s(t_i)| \le \epsilon$ so $z \in D_i$. For $w = \gamma_{s'}(t)$, $|w - z_i| = |\gamma_{s'}(t) - \gamma_s(t_i)| < \epsilon$.*

Finally, let F_i be a primitive of f on D_i (exists by Cauchy applied to D_i). On $D_i \cap D_{i+1}$, F_i, F_{i+1} are two primitives of the same function f so

$$F_{i+1}(w) - F_{i+1}(z) = c_i.$$

where c_i is a constant. Then

$$\gamma_{\gamma_s} f = \int_a^b f(\gamma_s(t))\gamma_s'(t)dt = \sum_{i=0}^{n} \int_{t_i}^{t_{i+1}} f(\gamma_s(t))\gamma_s'(t)dt = \sum_{i=0}^{n} F_i(z_{i+1}) - F_i(z_i).$$

So

$$\begin{aligned}
\int_{\gamma_s} f - \int_{\gamma_{s'}} f &= \sum_{i=0}^{n} [F_i(z_{i+1}) - F_i(z_i)] - \sum_{i=0}^{n} [F_{i+1}(w_{i+1}) - F_{i+1}(w_i)] \\
&= \sum_{i=0}^{n} [F_i(z_{i+1}) - F_{i+1}(w_{i+1})] - \sum_{i=0}^{n} [F_i(z_i) - F_{i+1}(w_i)] \\
&= F_0(w_0) - F_0(z_0) + \sum_{i=0}^{n-1} [F_i(z_{i+1}) - F_{i+1}(w_{i+1})] - \sum_{i=0}^{n-1} [F_i(z_i) - F_{i+1}(w_i)] \\
&= \sum_{i=0}^{n-1} [F_i(z_{i+1}) - F_{i+1}(w_{i+1})] - \sum_{i=0}^{n-1} [F_i(z_i) - F_{i+1}(w_i)] \\
&= \sum_{i=0}^{n-1} [F_i(z_{i+1}) - F_{i+1}(z_i) + F_{i+1}(z_i) - F_i(w_{i+1})] \\
&= \sum_{i=0}^{n-1} -c_i + c_i = 0
\end{aligned}$$

□

16.2 Simply Connected Regions

17 Feb 23 — Complex Logarithm and General Powers

17.1 Complex Logarithm

Theorem 17.1. *(Construction of Complex Logarithm) Let Ω be a simply connected region such that $0 \notin \Omega$ but $1 \in \Omega$. Then there $\exists!$ holomorphic F on Ω with $F(z) = \log \Omega$ such that*

1. $e^{F(z)} = z$ *on* Ω
2. $F(r) = \log r \ \forall r \in \Omega \cap (1-\delta, 1+\delta)$

Proof. Let $F(z) = \int_{\gamma_z} \frac{dw}{w}$ where γ_z is a curve $1 \curvearrowright z$ in Ω. This is well defined as it doesn't depend on choice of γ_z because Ω is simply connected.
Then we know $F'(z) = \frac{1}{z}$.
Thus

1. $(ze^{-F(z)})' = e^{-F(z)} - ze^{-F(z)}\frac{1}{z} = 0$ so $ze^{-F(z)} = 1$ at $z = 1$.
2. $F(r) = \int_1^r \frac{dt}{t} = \log t \,|_{t=1}^{t=r} = \log r$

□

Example 17.2. $\Omega_0 = \mathbb{C} \setminus (-\infty, 0]$

Claim 17.3. *For $z = re^{i\theta}$ for $|\theta| < \pi$, $\log(z) = \log r + i\theta$, we take the path $1 \to r \to z$. So*

$$\log z = \int_{\gamma_z} \frac{dw}{w} = \int_1^r \frac{dt}{t} + \int_0^{2\pi} \frac{rie^{it}}{re^{it}} dt = \log r + i\theta.$$

Remark 17.4. $\log(z_1 z_2) \neq \log z_1 + \log z_2$ *is in general false.*

Theorem 17.5. *For the principle branch*

$$\log(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \cdots. \qquad \text{(For } |z| < 1)$$

Proof. $\frac{d}{dz}(\text{LHS} - \text{RHS}) = \frac{1}{1+z} - \frac{1}{1+z} = 0$ □

17.2 General Powers and Branches

Definition 17.6. *Given α*

18 Feb 25 — Fourier Transform and Paley-Wiener Theorem

18.1 Fourier Transform and Basic Properties

Definition 18.1. *For $f \in L_1$ $(\mathbb{R} \to \mathbb{C})$ i.e. $||f||_{L_1} = \int_{-\infty}^{+\infty} |f| < \infty$ we define its fourier transform*

$$\hat{f}(\zeta) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i t \zeta} dt. \qquad (\zeta \in \mathbb{R})$$

Example 18.2. *Some examples of fourier transforms:*

1. $\widehat{e^{-\pi x^2}}(\zeta) = e^{-\pi \zeta^2}$
2. $\widehat{\frac{1}{\cosh(\pi x)}}(\zeta) = \frac{1}{\cosh(\pi \zeta)}$
3. $\widehat{1_{[-1,1]}}(\zeta) = \int_{-1}^{1} e^{-2\pi i \zeta t} dt$

Lemma 18.3. *Basic Properties of Fourier Transforms:*

1. *$\hat{f}$ is uniformly continuous*
2. *$||\hat{f}||_\infty \leq ||f||_{L_1}$*
3. *$\hat{f}(\zeta) \to 0$ as $|\zeta| \to \infty$*

Theorem 18.4. *(Paley-Wiener Theorem) Regularity of f $\iff$ decay of f^**

18.2 Paley-Wiener Theorem in a Strip

Definition 18.5. *Given $a > 0$, we let $S_a = \{z \in \mathbb{C} : |\Im(z)| < a\}$ and*

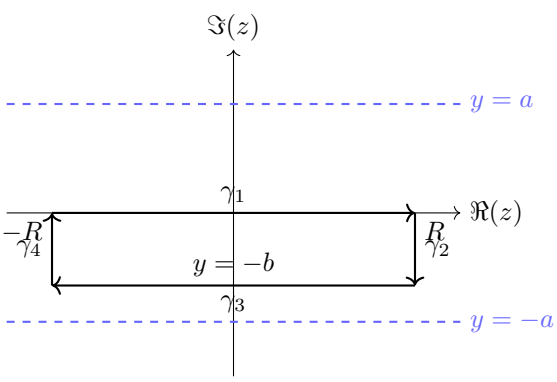
$$\mathcal{F}_a = \left\{ f : S_a \to \mathbb{C},\ holo\ , \exists C_f, \forall x + iy \in S_a, |f(x+iy)| \leq \frac{C_f}{1+x^2} \right\}.$$

Additionally, $\mathcal{F} = \bigcup_{a>0} \mathcal{F}_a$

Theorem 18.6. *If $f \in \mathcal{F}_a$, then $|\hat{f}(\zeta)| \leq Ce^{-2\pi b|\zeta|}$, for $\zeta \in \mathbb{R}$ and $\forall b < a$.*

Proof. Fix $\zeta > 0$ and $b < a$, we have

$$0 = \int_{\gamma_R} f(z) e^{-2\pi i \zeta z} dz = \int_{\gamma_1} + \int_{\gamma_2} + \int_{\gamma_3} + \int_{\gamma_4}.$$



Verify that $\int_{\gamma_1} = \hat{f}(\zeta)$ and

$$\begin{aligned}\left|\int_{\gamma_2}\right| &= \left|\int_0^b f(R-it)e^{-2\pi i\zeta(R-it)}(-i)\,dt\right| \\ &\leq \int_0^b |f(R-it)|e^{-2\pi\zeta t}dt \\ &\leq \frac{C_f}{1+R^2}\int_0^b e^{-2\pi\zeta t}dt \to 0 \text{ as } R\to\infty\end{aligned}$$

Additionally,

$$\left|\int_{\gamma_4}\right| \to 0.$$

Thus

$$\hat{f}(\zeta) = \int_{-\infty}^{\infty} f(t-ib)e^{-2\pi i\zeta(t-ib)}dt.$$

and

$$\left|\hat{f}(\zeta)\right| \leq \int_{-\infty}^{\infty} |f(t-ib)|e^{-2\pi\zeta b}dt \leq \int_{-\infty}^{\infty} \frac{C_f}{1+t^2}e^{-2\pi\zeta b}dt = C_f e^{-2\pi b\zeta}\left(\int_{-\infty}^{\infty}\frac{dt}{1+t^2}\right). \qquad (\zeta > 0)$$

Do same for $\zeta < 0$. □

Theorem 18.7. *(Inversion Formula) For $f \in \mathcal{F}_\infty$,*

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\zeta)e^{2\pi i\zeta x}d\zeta. \qquad (x \in \mathbb{R})$$

Proof. We want to show that $g(x) = \int_{-\infty}^{\infty} \hat{f}(\zeta)e^{2\pi i\zeta x}d\zeta = f(x)$ on $\mathbb{R}$.
We can write

$$\int_0^{\infty} \hat{f}(\zeta)e^{2\pi i\zeta x}d\zeta = \int_{ib+\mathbb{R}} \cdots .$$

Rewriting we have

$$\begin{aligned}\int_0^{\infty} \hat{f}(\zeta)e^{2\pi i\zeta x}d\zeta &= \int_0^{\infty}\left(\int_{-\infty}^{\infty} f(t-ib)e^{2\pi i\zeta(t-ib)}dt\right)e^{2\pi i\zeta x}d\zeta \\ &= \int_{-\infty}^{\infty} f(t-ib)\left(\int_0^{\infty} e^{-2\pi\zeta(b+it-ix)}d\zeta\right.\end{aligned}$$

19 March 9 — Paley-Wiener Equivalence and Analytic Extension

19.1 Paley-Wiener Equivalence and Fourier Inversion

Theorem 19.1. *$f \in \mathcal{F}_a$ iff $|\hat{f}(\zeta)| \leq Ce^{-2\pi b|\zeta|}$, for $\zeta \in \mathbb{R}$ and $\forall b < a$.*

Proof. We've shown the forward direction in Theorem 18.6 □

Theorem 19.2. *If $f : \mathbb{R} \to \mathbb{C}$ continuous with $|f(x)| \leq \frac{C_f}{1+x^2}$ for all $x \in \mathbb{R}$ and $\hat{f} \in L_1$ then Fourier Inversion Formula holds.*

Proof. Shown on the homework. □

19.2 Analytic Extension and Uncertainty Principle

Theorem 19.3. *If $|\hat{f}(\zeta)| \leq Ce^{-2\pi a|\zeta|}$, $\zeta \in \mathbb{R}$ and $\forall a > 0$ then f extends analytically to $S_a = \{|\Im(z)| < a\}$.*

Proof. Let

$$f_n(z) = \int_{-n}^{n} \hat{f}(\zeta)e^{2\pi i\zeta z}d\zeta.$$

Fix $b < a$ then

$$\tilde{f}(z) = \int_{-\infty}^{\infty} \hat{f}(\zeta)e^{2\pi i\zeta z}d\zeta \text{ is well defined in } S_b.$$

By Fourier Inversion $\tilde{f}|_{\mathbb{R}} = f$, it remains to show f_n converges uniformly to $\tilde{f}$ on S_b.

$$\begin{aligned} \left|\tilde{f}(z) - f_n(z)\right| &= \left|\int_{|\zeta|>n} \hat{f}(\zeta)e^{2\pi i\zeta z}d\zeta\right| \\ &\leq \int_{|\zeta|>n} Ce^{-2\pi a|\zeta|+2\pi|\zeta|b}d\zeta \end{aligned} \tag{1}$$

Using

$$\left|e^{2\pi i\zeta z}\right| = e^{\Re(2\pi i\zeta z)} = e^{-2\pi\zeta y} \leq e^{2\pi|\zeta|b}.$$

We have (1) $\to 0$ as $n \to +\infty$ □

Corollary 19.4. *(Baby Uncertainty Principle) If $\hat{f}(\zeta) = O(e^{-c|\zeta|})$ and f vanishes along a sequence without limit point then $f \equiv 0$.*

20 March 11 — Paley-Wiener Theorem for Compact Support

20.1 Support and Growth of Entire Functions

Theorem 20.1. *Theorem 2 of Paley-Wiener: $f : \mathbb{R} \to \mathbb{C}$ continuous then f extends to an entire function with $|f(z)| \leq Ae^{2\pi M|z|}$ for all $z \in \mathbb{C}$ iff $\{z \in \mathbb{C} : \hat{f}(\zeta) \neq 0\} = supp(\hat{f}) \subset [-M, M]$.*

Proof. ($\Longleftarrow$) By the inversion formula

$$f(x) = \int_{-M}^{M} \hat{f}(\zeta) e^{2\pi i \zeta x} d\zeta, \ x \in \mathbb{R}.$$

This is a function of x and entire. For $z = x + iy$

$$|f(z)| \leq \int_{-M}^{M} |\hat{f}(\zeta)| e^{-2\pi\zeta y} d\zeta \leq \int_{-M}^{M} |\hat{f}(\zeta)| e^{2\pi M|y|} d\zeta \leq \int_{-M}^{M} |\hat{f}| e^{2\pi M|z|} d\zeta.$$

($\Longrightarrow$)
Step 1:
Assume that

$$|f(x+iy)| \leq A \frac{e^{2\pi M|x+iy|}}{1+x^2}. \tag{$*$}$$

Then, for $\zeta > M$,

$$\hat{f}(\zeta) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i \zeta x} dx.$$

If we take the contour integral of the rectangle, then the vertical parts go to 0 and we are left with the horizontal piece so for all $y > 0$

$$\hat{f}(\zeta) = \int_{-\infty}^{\infty} f(x - iy) e^{-2\pi i \zeta (x - iy)} dx$$

Upper bounding using $(*)$ we have

$$|\hat{f}(\zeta)| \leq \int_{-\infty}^{\infty} A \frac{e^{2\pi M y}}{1+x^2} e^{-2\pi\zeta y} dx = (\pi A) e^{-2\pi y(\zeta - M)} \xrightarrow{y \to \infty} 0. \tag{$\star$}$$

Step 2:
Assume

$$|f(x+iy)| \leq Ae^{2\pi M|y|}, \ \forall x, y \in \mathbb{R}. \tag{$**$}$$

Similarly, consider $\hat{f}(\zeta) = 0$ for $\zeta < -M$ by shifting the contour up.
At this point we've only used that f is holomorphic on $\{\Im(z) < 0\}$.
For $\epsilon > 0$, consider

$$f_\epsilon(z) = \frac{f(z)}{(1 + i\epsilon z)^2} \text{ poles at } \frac{1}{-i\epsilon} = \frac{i}{\epsilon}.$$

For $x \in \mathbb{R}, y > 0$, we have

$$|f_\epsilon(x - iy)| \leq \frac{|f(x - iy)|}{\epsilon^2 x^2 + (1 + \epsilon y)^2} \leq \frac{Ae^{2\pi M|y|}}{\epsilon^2(1 + x^2)}.$$

So f_ϵ satisfies $(*)$, so by $(\star)$ applied to f_ϵ gives $\hat{f}_\epsilon(\zeta) = 0, \zeta > M$.
Now

$$\left|\hat{f}_\epsilon(\zeta) - \hat{f}(\zeta)\right| = \left|\int_{-\infty}^{\infty} (f_\epsilon(x) - f(x)) e^{-2\pi i \zeta x} dx\right| = \left|\int_{-\infty}^{\infty} \left(\frac{1}{(1 + i\epsilon x)^2} - 1\right) f(x) e^{-2\pi i \zeta x} dx\right|.$$

Goes to 0 by Lebesgue Dominated Convergence Theorem.
Step 3:
Claim: If f entire and

$$|f(z)| \leq Ae^{2\pi M|z|}, \ \forall z \in \mathbb{C}. \tag{$***$}$$

then $(**)$ holds.
Let's assume $|f(x)| \leq 1$ on $x \in \mathbb{R}$, $|f(z)| \leq e^{2\pi M|z|}$ and $Q = \{\Re z | \Im(z) > 0\}$. Lemma: For $F(z) = f(z)e^{2\pi i M z}$ on ∂Q:

$$|F(z)| = |f(x)||e^{2\pi i M x}| \leq 1.$$

$$|F(iy)| \leq e^{2\pi M y} e^{-2\pi M y} = 1.$$

On Q plainly, $|F(z)| \leq Ce^{c'|z|}$. Thus $|F| \leq 1$ on Q.
If $z = x + iy \in Q$,

$$|f(x+iy)||e^{2\pi i M x} \cdot e^{-2\pi M y}| = |F(x+iy)| \leq 1.$$

So $|f(x+iy)| \leq e^{2\pi M y}$ on Q. Similarly for the other 3 quadrants of $\mathbb{C}$. $\square$

21 March 13 — Phragmen-Lindelof and Conformal Maps

21.1 Phragmen-Lindelof Principle

Lemma 21.1. *(Phragmen-Lindelof Principle) Let $S = \{z \in \mathbb{C} : |\arg(z)| < \frac{\pi}{4}\}$. Let F be holomorphic in S and continuous on $\overline{S}$ and*

$$\begin{cases} |F(z)| \leq 1 & \text{on } \partial S \\ |F(z)| \leq Ce^{c|z|} & \forall z \in S \end{cases}.$$

then $|F| \leq 1$ on S.

Proof. Idea: Turn our foe e^{z^2} into our friend, $\epsilon > 0$. Define $F_\epsilon(z) = F(z)\exp(-\epsilon z^{3/2})$. We have that $F_\epsilon(z) \xrightarrow{\epsilon \to 0} F(z)$.
On S: For $z = re^{i\theta}$, $|\theta| < \frac{\pi}{4}$,

$$|F_\epsilon(z)| \leq Ce^{cr}|\exp\left(-\epsilon r^{3/2}e^{\frac{3}{2}i\theta}\right)| = Ce^{cr}\exp\left(-\epsilon r^{3/2}\cos\left(\frac{3}{2}\theta\right)\right) \leq Ce^{cr-\epsilon\delta r^{3/2}} \xrightarrow{r \to +\infty} 0.$$

For $0 < \delta < \cos\left(\frac{3\pi}{8}\right)$ (so $\delta < \cos\left(\frac{3}{2}\theta\right)$ for all $|\theta| < \frac{\pi}{4}$), we have $|F_\epsilon| \xrightarrow{|z| \to +\infty} 0$.
So F_ϵ is bounded on S.
Claim: $|F_\epsilon| \leq 1$ on S.
Let $M = \sup_{\overline{S}} |F_\epsilon|$, if $M = 0$ then $F \equiv 0$ otherwise if $M > 0$ take $\overline{S} \supset w_1, w_2, \ldots$ such that $|F_\epsilon(w_j)| \xrightarrow{j \to +\infty} M$.
This sequence is bounded so wlog say $w_j \to w$. We have that $w \notin S$ as the open mapping theorem applied to F_ϵ.

$$1 \geq |F(w)| \cdot |\exp(-\epsilon w^{3/2}) = |F_\epsilon(w)| = M.$$

So $M \geq 1$ □

21.2 Conformal Maps

Given $U, V \subset \mathbb{C}$, when $\exists f : U \to V$ bijective holomorphic.

Lemma 21.2. *(Shown in HW) The only entire function which are injective are linear*

Theorem 21.3. *Riemann's Theorem: Let $\Omega \subset \mathbb{C}$ be open. Then $\exists f : \Omega \to \mathbb{D}$ bijective and holomorphic $\iff$ $\Omega \neq \mathbb{C} \& \Omega$ simply connected.*

Lemma 21.4. *Let U and V be open, $f : U \to V$ holomorphic with f' nonvanishing then f is a local bijection.*

Proof. Fix $z_0 \in U$, we want to find $\epsilon > 0$ such that $f|_{D_\epsilon(z_0)}$ is a bijection.
Near $f(z) - f(z_0) = a(z - z_0) + G(z)$ where $a = f'(z_0) \neq 0$. Consider small w then $f(z) - f(z_0) - w = \underbrace{a(z - z_0) - w}_{F(z)} + G(z)$.
On $\partial D_\epsilon(z_0)$:

$$|F(z)| \geq |a|\epsilon|w| \geq C\epsilon^2 \geq |G(z)|.$$

provided that ϵ small and all sufficiently small w.
By Rouche's: F and $F + G$ have the same number of zeros in $D_\epsilon(z_0)$ but F has exactly one zero. □

22 March 16 — Injectivity and the Derivative

22.1 Injectivity and Nonvanishing Derivative

Theorem 22.1. *$f : U \to V$ holomorphic, f is 1-1, then f' never vanishes.*